ERSHAD KUNNAKKADAN

Staff Software Engineer | Former CEO, Gumroad

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Summary

Staff software engineer with 14 years of experience and former CEO of Gumroad. Recognized by the U.S. government as an individual with extraordinary ability (O-1A). Lead for Gumroad's infrastructure, security and performance since 2017, now at 7B+ requests a month. After serving as CEO, chose to return to hands-on engineering to build AI-native products and agent-driven systems.

Experience

Gumroad Jun 2020 – Present

New York, NY and remote. Open-source creator-commerce platform with $100M+ annual GMV and 7B+ monthly requests.

Staff Software Engineer *Mar 2026 – Present*

- Migrated email analytics from MongoDB to Amazon DynamoDB: backfilled ~970M items with zero mismatches, cut cost 88%, and sped up stats lookups from 861 ms to 7.3 ms.
- Cut AWS spend by ~$100K a year in two weeks, including a MySQL 8.4 upgrade ending ~$39K/year in Extended Support fees and a Sidekiq redesign cutting p99 queue wait 74 s → 19 s at 69% lower cost.
- Fixed the root causes of recurring 502/504 incidents; 504 bursts during deploys fell from 6K–17K to zero.
- Upgraded the AWS fleet to Debian 12 with no disruption, moved to AMD instances (~10% savings), and halved Docker image size.
- Hardened security: TOTP 2FA, webhook signature checks, SSRF-safe redirects, checkout reCAPTCHA.
- Built AI-native internal tools with Claude Code at ~3.5x the 2025 shipping pace, and supervised Gumclaw, the company's autonomous ops agent.
- Security-audited Flexile's agent-written Rails migration before launch, fixing ~15 vulnerabilities, including an account-takeover path.

Chief Executive Officer *Nov 2025 – Mar 2026*

- Led strategy, product and operations for Gumroad, Flexile and Small Bets; introduced quarterly OKRs.
- Oversaw operations delivering $459K net income in Jan 2026 (+11.7% YoY) and $859K in Feb 2026.
- Raised shipping velocity to 155 merged PRs in Feb 2026, finished the Inertia.js and Tailwind CSS migrations, and started the Expo mobile app; personally merged or reviewed nearly 140 PRs.
- Hired engineers and customer support staff to expand the team, and presented the live-streamed 2026 public company meeting.
- Grew the open-source community with live-streamed PR reviews, a paid OSS-reviewer program and a bounties board; drove fixes for 8 critical audit findings.

Staff Software Engineer *Jan 2025 – Nov 2025*

- Led the technical work to open-source Gumroad: scrubbed secrets from code and 3,261 test fixtures, authored the 9,680-file initial public commit, and made tests run without internal credentials. Did the same for Helper and Flexile.
- Fixed 15 findings from a third-party security audit before launch (SSRF, CSRF, IDOR, privilege escalation, XSS, timing attacks).

- Replaced Clerk with in-house OTP/JWT authentication in Flexile, and added automatic payout pauses when a seller's payout failure rate spikes.
- Created Gumboard, an open-source team to-do board, and built its MVP solo.

Senior Software Engineer (Infrastructure, Security & Performance Lead) *Jun 2020 – Dec 2024*

- Primary owner of Gumroad's AWS infrastructure (Terraform, Nomad, Consul, Vault, Packer) with 82% of its commits; upgraded Terraform 0.11 → 1.2 across 230 files with zero drift.
- Moved CI from Buildkite/EC2 to 128 self-hosted GitHub Actions runners on Hetzner and Ubicloud, for a projected 90% cut in CI spend.
- Stopped a distributed brute-force attack on the OAuth endpoint with Rack::Attack and Nginx rate limits, then built 2FA and enabled it for every user.
- Built automated fraud and card-testing defenses (velocity limits, card and PayPal blocklists, reCAPTCHA, auto-suspension) to curb ~$100K/year in chargeback fees.
- Core engineer on Helper v1, Antiwork's AI customer-support product, from prototype to paid SaaS: GPT-4 drafting with RAG, LLM triage, Stripe usage billing, and encryption at rest.
- Migrated video transcoding to AWS MediaConvert; re-architected creator subdomains.

BigBinary Jul 2015 – Jun 2020

Remote. Ruby on Rails consultancy; embedded with Gumroad as infrastructure lead from 2017.

Senior Software Engineer *Jul 2015 – Jun 2020*

- Moved Gumroad's production from Capistrano/EC2 to Docker on Nomad and Consul with zero-downtime blue/green deploys; moved secrets into Vault.
- Built per-branch preview environments, auto-deployed and posted to Slack, used for all QA.
- Automated S3 cleanup that deleted 154 TB of stale data; added free SSL for creators' custom domains.
- Delivered Rails and DevOps work for other clients; wrote 10 BigBinary blog posts.

Bang the Table Mar 2012 – Jul 2015

Bangalore, India

Software Architect *Jan 2013 – Jul 2015*

- Led an engineering team on EngagementHQ, the flagship product; owned DevOps and security.

Software Engineer (from Developer Intern) *Mar 2012 – Dec 2012*

Skills

Languages: Ruby, Python, TypeScript, JavaScript, Go, Swift, SQL, Bash, Lua

Backend: Ruby on Rails, Sidekiq, Puma, RSpec, Inertia.js, Django

Architecture: High-traffic systems, background jobs, autoscaling, zero-downtime deploys, data migrations

Cloud & Infrastructure: AWS (EC2, ALB, Auto Scaling, RDS, S3 / Glacier, CloudFront, Route 53, IAM, VPC, CloudWatch, cost optimization, Lambda, ECR, MediaConvert, SNS, SQS, SSM), Cloudflare, Terraform, Packer, Nomad, Consul, Vault, Docker, Nginx / OpenResty, HAProxy, Heroku, Vercel, Hetzner, Ubicloud, Ansible, Linux

CI/CD & Observability: GitHub Actions, Buildkite, CircleCI, Sentry, New Relic

Data Stores: MySQL (RDS), PostgreSQL, DynamoDB, Redis (ElastiCache), MongoDB, Elasticsearch, Metabase

AI & LLMs: Claude Code, Anthropic API, OpenAI API, RAG, AI agents and agent operations

Security: Application security (SSRF, XSS, CSRF, IDOR), 2FA / TOTP, OAuth, rate limiting, fraud and card-testing prevention, secrets management, encryption at rest, Let's Encrypt / ACME

Frontend & Mobile: React, Next.js, Tailwind CSS, Vite, Expo / React Native

Payments & Messaging: Stripe (Connect, Billing), PayPal, Braintree, Wise, SendGrid, Resend, Slack API

Education

Bachelor's degree in Computer Science | Bharathiar University, India

Security Research

- **GitHub Security Bug Bounty**: found a vulnerability involving forked private Gists (bounty.github.com/researchers/ershad.html).
- **Prezi Bug Bounty**: found a vulnerability that exposed the titles of other users' private presentations.

Open Source

- **Google Summer of Code**: mentor (2014, Diaspora language detection); student (2013, Grandham).
- **Swathanthra Malayalam Computing** (2009 – Present): free-software development and localization.
- **GitLab** (2017): built user mention notifications in issue descriptions and comments.
- **Wikimedia** (2012): fixed issues in the Universal Language Selector.

Awards & Talks

- Won "Most Innovative Project" at Wikimedia DevCamp, IIM Bangalore (2012).
- Won 9 first prizes in national programming, shell-scripting, security and bot contests (2011–2012).
- Gave 12 talks and workshops on Ruby, Rails, Git and Linux, including at Rails Girls Kochi (2018).